Difficulty: Medium

Problem Statement: *"BattleQuest – Character System"*

You are developing a **role-playing game (RPG)** called **BattleQuest**.
In the game, different types of characters fight using different abilities.

You must design the system using **all types of inheritance** and **polymorphism** in Java.

□ Requirements

1 Base Class (Single Inheritance – Parent)

Create a class **`GameCharacter`** with:

Fields

- `String name`
- `int health`
- `int level`

Methods

- `void attack()` → prints a general attack message
- `void defend()` → prints a general defend message
- `void displayInfo()` → prints character details

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2 Derived Classes (Hierarchical Inheritance)

Create the following subclasses of `GameCharacter`:

□□ **`Warrior`**

- Additional field: `int strength`
- Override `attack()` → Warrior attacks with sword
- Add method: `void powerStrike()`

□ **`Mage`**

- Additional field: `int mana`
- Override `attack()` → Mage casts a spell
- Add method: `void castSpell()`

 `Archer`

- Additional field: `int arrows`
- Override `attack()` → Archer shoots arrows
- Add method: `void rapidShot()`

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3 Multilevel Inheritance

Create a class **`EliteWarrior`** that extends `Warrior`:

Additional field

- `String specialWeapon`

Methods

- Override `attack()` → Elite warrior attacks using special weapon
- Add method: `void ultimateMove()`

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4 Method Overloading (Compile-time Polymorphism)

In **`GameCharacter`**, overload the `attack()` method:

- `void attack()`
- `void attack(String weapon)`
- `void attack(int powerLevel)`

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5 Method Overriding (Runtime Polymorphism)

- Override `attack()` in all child classes.
- Use **dynamic method dispatch** using parent reference.

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6 Upcasting & Polymorphism

In the main class:

- Store different character objects (`Warrior`, `Mage`, `Archer`, `EliteWarrior`) inside a `GameCharacter[]` array.
- Call `attack()` and `defend()` using the parent reference.

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7 Final Task (Main Class)

Create a class **`BattleQuestGame`** with `main()` method where:

- Create at least **4 characters**
- Demonstrate:
  - All types of inheritance
  - Method overloading
  - Method overriding
  - Runtime polymorphism
- Print meaningful output for gameplay

Expected Output (Sample – Not Fixed)

```
Warrior Arjun attacks with sword!
Mage Ishani casts a fire spell!
Archer Veer shoots arrows!
Elite Warrior Raghav attacks using Dragon Blade!

All characters defend!
```